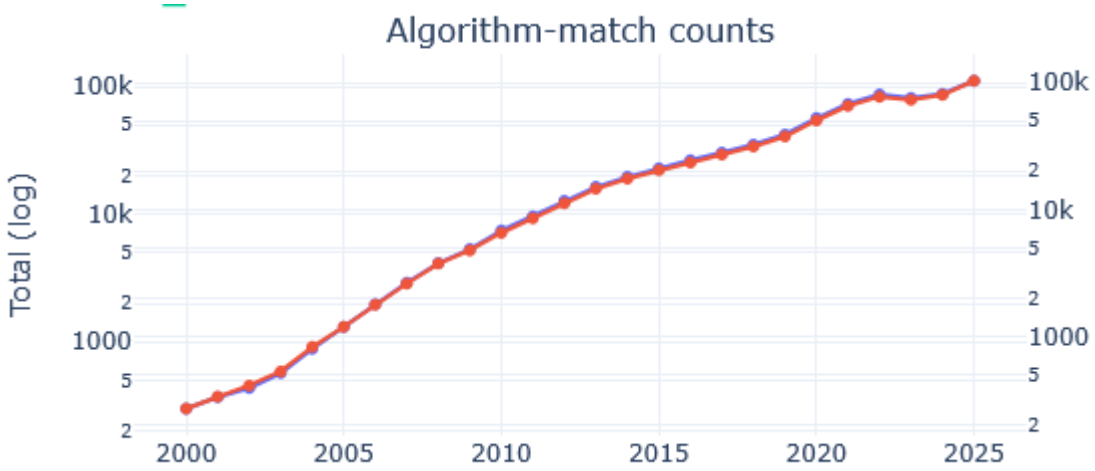
Results

- For each article, we run all regex in DEFAULT_PATTERNS["algorithm"] on the article body text.
- If at least one of those regex matches anywhere in the text, then that article is counted as having an algorithm match.
- The yearly metric is the number of such articles in that year.

$$\text{articles_with_any_algorithm_match}_y = \sum_{\text{articles in year } y} \mathbf{1}[\text{algorithms_found} \neq \emptyset]$$



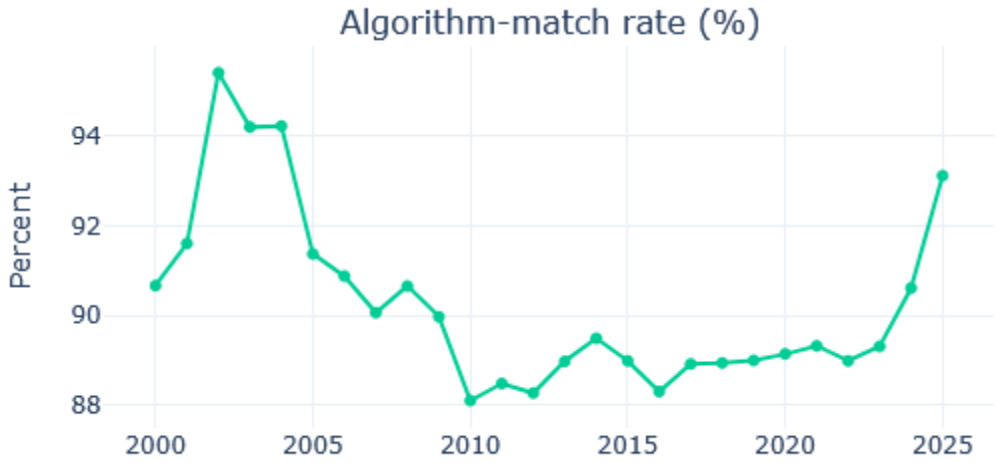
RapCluster

Chart **1**

Results

- the percent of articles in that year where at least one regex in DEFAULT_PATTERNS["algorithm"] matched the article body text.

$$\text{pct_with_any_algorithm_match}_y = 100 \times \frac{\text{articles_with_any_algorithm_match}_y}{\text{total_articles}_y}$$



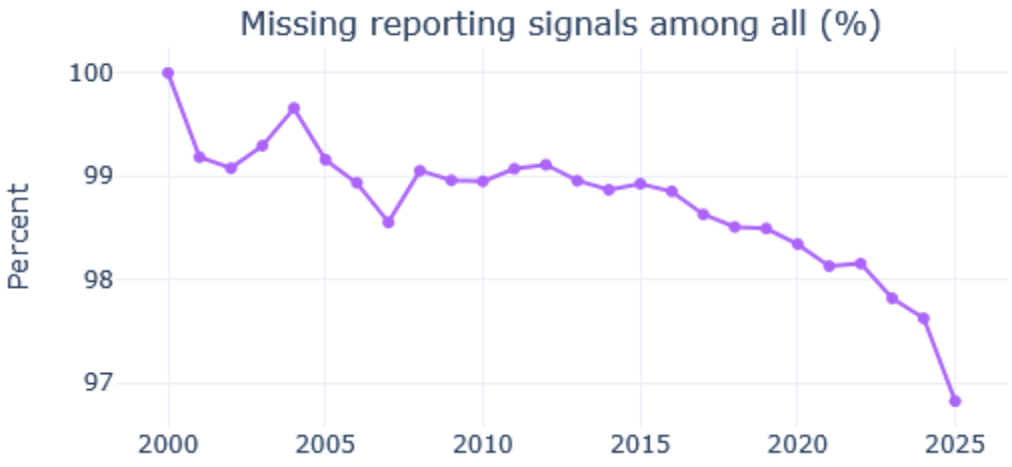
RapCluster

Chart **2**

Results

- This is an article-level flag that becomes 1 when the article is missing at least one of the four "reporting signal" categories.

$$\text{missing_reporting_signals}_i = \mathbf{1}\left(\text{missing_params}_i \vee \text{missing_justification}_i \vee \text{missing_evaluation}_i \vee \text{missing_tuning}_i\right)$$



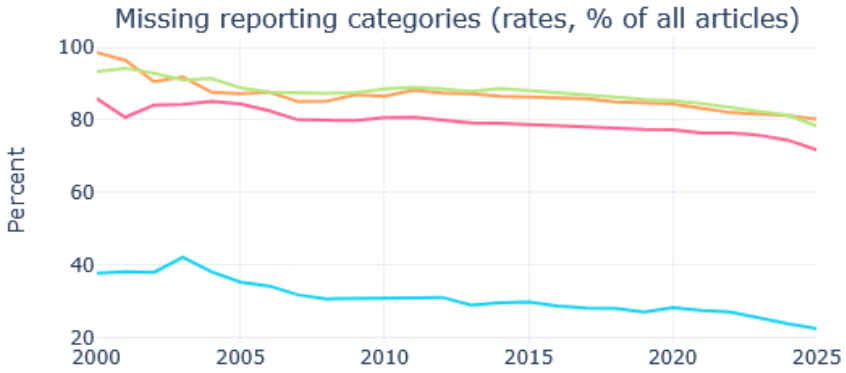
RapCluster

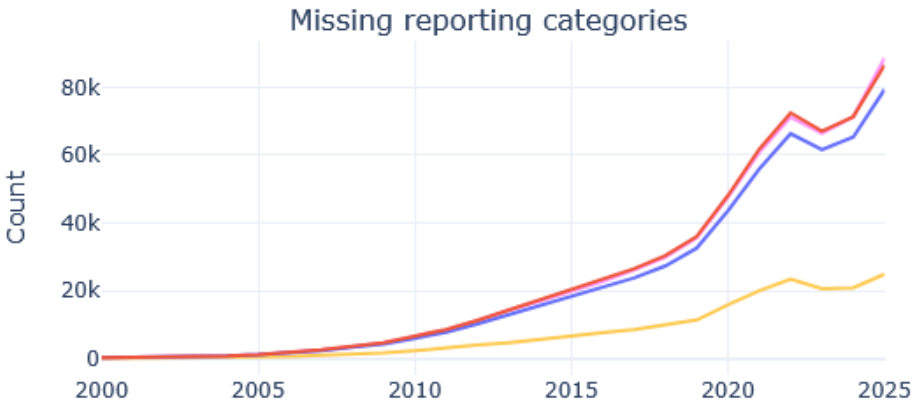
Chart **3**

Results

- **missing_params:** no explicit parameter value/config mentioned (k, eps, resolution, n_clusters, linkage, etc.).
- **missing_justification:** no explicit reasoning/selection rationale for the parameter/cluster choice (elbow, silhouette analysis, “we chose… based on…”, etc.).
- **missing_evaluation:** no explicit validation/quality check of clustering (silhouette/DBI/CH, ARI/NMI, stability, BIC/AIC, modularity, etc.).
- **missing_tuning:** no explicit search/optimization process over parameters (grid/random search, Bayesian opt, CV, “tuned/optimized using…”, etc.).

Results





RapCluster

Chart **5**

Results

- Within a given year, how the total missingness flags are split across the four categories.

Concretely, for a year y:

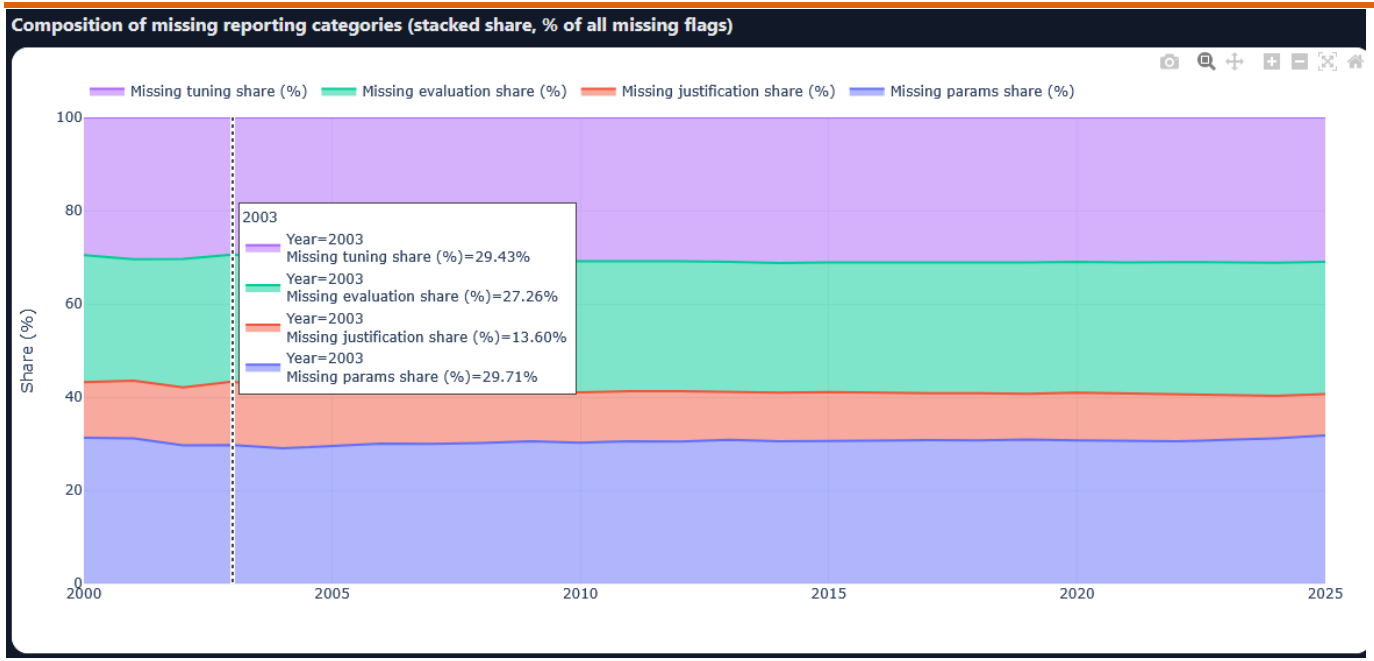
$$S_y = \text{missing_params}_y + \text{missing_justification}_y + \text{missing_evaluation}_y + \text{missing_tuning}_y$$

$$\text{missing_tuning_share}_y = 100 \times \frac{\text{missing_tuning}_y}{S_y}$$

missing_tuning_share (%) = "Of all missing-flags counted in that year (across the 4 categories), what fraction are tuning?"

RapCluster

Results



RapCluster

Chart **7**

Is there an effect of the algorithm used?

- When algorithm X is mentioned, do papers report params/justification/evaluation/tuning differently than the overall baseline?
- Computes the algorithm-specific missing rates for each missing flag.
- Computes the overall baseline missing rates (across all relevant articles; typically “all articles with any algorithm match”, sometimes “all articles overall” depending on how the script was written).
- Reports effect sizes like deltas vs overall, and often statistical tests.